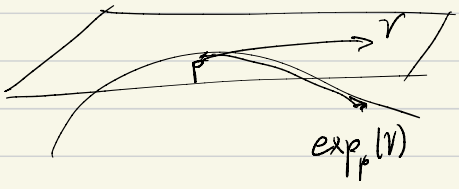
Exponential map

define $\exp : \mathcal{U} \subset TM \longrightarrow M$ by

$V \in T_pM \longmapsto \exp(V) = \gamma_V(1)$ where γ_V is the geodesic with $\gamma_V(0) = p$, $\dot\gamma_V(0) = V$

($V \in \mathcal{U}$ if γ_V exists at $t = 1$)



Rescaling Lemma: If γ is a geodesic of constant speed $r = \|\dot\gamma\|$ then $\eta(t) := \gamma(ct)$ is a geodesic of speed cr

Proof: The chain rule gives $\|\dot\eta\| = c\|\dot\gamma\|$

Check $\nabla_{\dot\eta}\dot\eta = c^2 \nabla_{\dot\gamma}\dot\gamma$

Cor $\gamma_V(t) = \exp(tV)$ as long as both sides are defined

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Prop.: $\mathcal{U}$ is an open subset of the 0-section in TM and exp is a smooth map.

The proof is an ODE regularity thm:

$\dot u^j = f^j(t, u)$ where $f^j \in C^\infty(\mathbb{R}^{m+1})$ $\quad j = 1, \dots, m$

$u^j(0) = a^j \qquad j = 1, \dots m$

$\longrightarrow$ unique solution $u(t; a)$ is smooth as a function of $(t; a) \in \mathbb{R}^{m+1}$

The term "exponential" comes from its differential

Since $\exp_p : T_pM \longrightarrow M$

$d(\exp)_X$ maps $T_X(T_pM) \longrightarrow T_{\exp_p(X)}M$

Since T_pM is a vector space, $T_X(T_pM) \cong T_pM$

This identification means that

$$d(\exp_p)_X V := \frac{d}{dt} \exp_p(X + tV)\Big|_{t=0}$$

In particular, $d(\exp_p)_0 V = \frac{d}{dt} \exp_p(tV)\Big|_{t=0}$

$$= \dot{\gamma}_V(0)$$

$$= V$$

<u>Lemma:</u> $d(\exp_p)_0 = I$ (identity map on T_pM)

Inverse fn thm: for $f : U_1 \longrightarrow V_1$ (open subsets of $\mathbb{R}^n$)
if $df(x)$ is invertible (linear map $\mathbb{R}^n \to \mathbb{R}^n$)
for some $x \in U_1$
Then $\exists$ nbhds $U \ni x$ and $V \ni f(x)$ so that
$f : U \to V$ is a diffeomorphism

<u>Cor</u> For each $p \in M$, $\exists$ nbhds $U \subset T_pM$ and $V \subset M$
so that

$$\exp_p : U \longrightarrow V \text{ is a diffeomorphism}$$

example: S^2 = unit sphere in $\mathbb{R}^3$

For $p \in S^2$ $\exp_p$ is diffeo as long
as we don't hit the antipode $-p$

$$\exp_p : \{|X| < \pi\} \longrightarrow S^2 \setminus \{-p\}$$

geodesic normal coordinates: Choose an orthonormal basis $\{E_j\}$ for T_pM

define $\mathbb{R}(x^1, \dots, x^n) = \exp_p(x^j E_j)$

The image of $\{|x| < r\}$ under $\mathbb{R}$ is called a **geodesic ball** $B_r(p) \subset M$

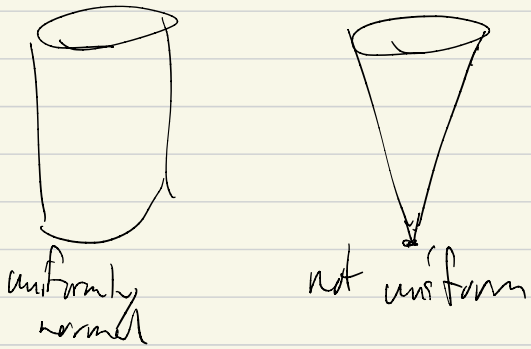
The radial curves in $B_r(p)$ are geodesics by the def. of exp.

Prop: In geodesic normal coord's,

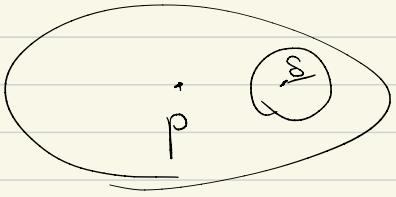
$$g_{ij}|_p = \delta_{ij}, \quad \partial_k g_{ij}|_p = 0$$

(trivial) (exercise)

Uniformity: a nbhd. $W \subset M$ is **uniformly normal** if $\exists \delta > 0$ so that each pt. has a geodesic ball of radius $\geq \delta$



Lemma Any pt. $p \in M$ admits a uniformly normal nbhd.



(The main idea is to compare, in normal coord's, the Euclidean norm to the g-norm.)

Geodesics and distance

A curve $\gamma : I \to M$ is <u>regular</u> if $\dot\gamma \neq 0$ for all t

($\Longrightarrow$ admits unit speed reparam.)

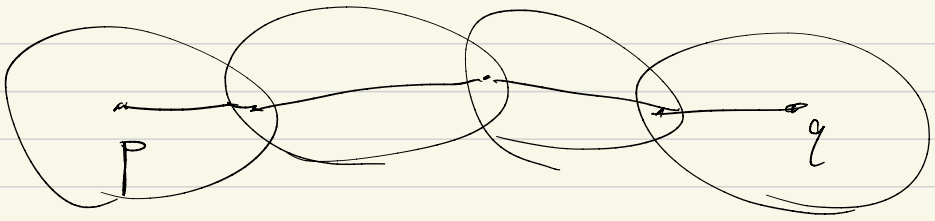
For regular curves $\gamma : [a,b] \to M$ define the length functional

$$L(\gamma) := \int_a^b \|\dot\gamma\| \, dt$$

This is easily extended to "piecewise" regular = continuous, finite union of regular segments



Given $p, q \in M$, there exists a pcwise regular path p to q given by joining regular paths from coordinate nbhds.



define the distance

$$d(p,q) = \inf\{L(\gamma) : \gamma \text{ pcwise regular } p \text{ to } q\}$$

<u>Lemma</u> This definition makes (M, g) a metric space, and the metric topology agrees with the manifold topology.